```
!pip install transformers
!pip install datasets
!pip install torch
!pip install scikit-learn
```

```
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.0.2)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.12/dist-packages (from datasets) (4.67.3)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025
Requirement already satisfied: huggingface-hub>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (1.5.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from datasets) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets) (6.0.3)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->data
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->datasets) (0.24
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->data
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.11
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fssp
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.24.
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hu
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->hug
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->datas
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.0->d
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.2
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.0->d
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->hug
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->h
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.0)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: numpy>=1.19.5 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (2.0.2)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
```

```
import torch
from datasets import load_dataset
from transformers import AutoTokenizer, AutoModelForSequenceClassification
from transformers import TrainingArguments, Trainer
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
```

```
dataset = load_dataset("glue", "sst2")

print(dataset)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
```

README.md: 35.3k/? [00:00<00:00, 2.33MB/s]

sst2/train-00000-of-00001.parquet: 100% 3.11M/3.11M [00:01<00:00, 7.07MB/s]

sst2/validation-00000-of-00001.parquet: 100% 72.8k/72.8k [00:00<00:00, 323kB/s]

sst2/test-00000-of-00001.parquet: 100% 148k/148k [00:00<00:00, 650kB/s]

Generating train split: 100% 67349/67349 [00:00<00:00, 295797.81 examples/s]

Generating validation split: 100% 872/872 [00:00<00:00, 50231.88 examples/s]

Generating test split: 100% 1821/1821 [00:00<00:00, 87861.82 examples/s]

```
DatasetDict({
    train: Dataset({
        features: ['sentence', 'label', 'idx'],
        num_rows: 67349
    })
    validation: Dataset({
        features: ['sentence', 'label', 'idx'],
        num_rows: 872
    })
    test: Dataset({
        features: ['sentence', 'label', 'idx'],
        num_rows: 1821
    })
})
```

```
dataset['train'][0]
```

```
{'sentence': 'hide new secretions from the parental units ',
 'label': 0,
 'idx': 0}
```

```
tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")
```

config.json: 100% 483/483 [00:00<00:00, 36.0kB/s]

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 3.84kB/s]

vocab.txt: 232k/? [00:00<00:00, 5.41MB/s]

tokenizer.json: 466k/? [00:00<00:00, 16.4MB/s]

```
def tokenize_function(example):
    return tokenizer(example["sentence"], padding="max_length", truncation=True)

tokenized_datasets = dataset.map(tokenize_function, batched=True)
```

Map: 100% 67349/67349 [00:28<00:00, 2616.72 examples/s]

Map: 100% 872/872 [00:00<00:00, 2284.27 examples/s]

Map: 100% 1821/1821 [00:00<00:00, 2456.03 examples/s]

```
train_dataset = tokenized_datasets["train"]
test_dataset = tokenized_datasets["validation"]
```

```
train_dataset = train_dataset.remove_columns(["sentence", "idx"])
test_dataset = test_dataset.remove_columns(["sentence", "idx"])
```

```
train_dataset.set_format("torch")
test_dataset.set_format("torch")
```

```
model = AutoModelForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
```

```
    num_labels=2
)
```

model.safetensors: 100% 268M/268M [00:05<00:00, 32.3MB/s]

Loading weights: 100% 100/100 [00:00<00:00, 420.29it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]

```
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                     | Status     |
------------------------+------------+-
vocab_transform.bias    | UNEXPECTED |
vocab_transform.weight  | UNEXPECTED |
vocab_layer_norm.bias   | UNEXPECTED |
vocab_projector.bias    | UNEXPECTED |
vocab_layer_norm.weight | UNEXPECTED |
pre_classifier.weight   | MISSING    |
pre_classifier.bias     | MISSING    |
classifier.weight       | MISSING    |
classifier.bias         | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
```

```
def compute_metrics(pred):
    labels = pred.label_ids
    preds = pred.predictions.argmax(-1)

    precision, recall, f1, _ = precision_recall_fscore_support(
        labels, preds, average='binary'
    )

    acc = accuracy_score(labels, preds)

    return {
        'accuracy': acc,
        'f1': f1,
        'precision': precision,
        'recall': recall
    }
```

```
training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=3,
    weight_decay=0.01,
)
```

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset,
    compute_metrics=compute_metrics
)
```

```
import torch
print(torch.cuda.is_available())
```

```
True
```

```
train_dataset = tokenized_datasets["train"].select(range(5000))
test_dataset = tokenized_datasets["validation"].select(range(1000))
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1172/4128277225.py in <cell line: 0>()
----> 1 train_dataset = tokenized_datasets["train"].select(range(5000))
      2 test_dataset = tokenized_datasets["validation"].select(range(1000))

NameError: name 'tokenized_datasets' is not defined
```

```
from datasets import load_dataset

dataset = load_dataset("glue", "sst2")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
```

README.md: 35.3k/? [00:00<00:00, 2.91MB/s]

sst2/train-00000-of-00001.parquet: 100% 3.11M/3.11M [00:02<00:00, 3.71MB/s]

sst2/validation-00000-of-00001.parquet: 100% 72.8k/72.8k [00:00<00:00, 345kB/s]

sst2/test-00000-of-00001.parquet: 100% 148k/148k [00:00<00:00, 741kB/s]

Generating train split: 100% 67349/67349 [00:00<00:00, 712721.51 examples/s]

Generating validation split: 100% 872/872 [00:00<00:00, 28909.54 examples/s]

Generating test split: 100% 1821/1821 [00:00<00:00, 67961.88 examples/s]

```
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")
```

config.json: 100% 483/483 [00:00<00:00, 53.8kB/s]

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 5.21kB/s]

vocab.txt: 232k/? [00:00<00:00, 2.84MB/s]

tokenizer.json: 466k/? [00:00<00:00, 7.25MB/s]

```
def tokenize_function(example):
    return tokenizer(example["sentence"], padding="max_length", truncation=True)

tokenized_datasets = dataset.map(tokenize_function, batched=True)
```

Map: 100% 67349/67349 [00:27<00:00, 2961.14 examples/s]

Map: 100% 872/872 [00:00<00:00, 2823.24 examples/s]

Map: 100% 1821/1821 [00:00<00:00, 2852.24 examples/s]

```
train_dataset = tokenized_datasets["train"].select(range(5000))
test_dataset = tokenized_datasets["validation"].select(range(tokenized_datasets["validation"].num_rows))
```

```
trainer.train()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1172/4032920361.py in <cell line: 0>()
----> 1 trainer.train()

NameError: name 'trainer' is not defined
```

```
!pip install transformers datasets torch scikit-learn
```

```
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in /usr/local/lib/python3.12/dist-packages (from torch) (10.3.9.9(
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in /usr/local/lib/python3.12/dist-packages (from torch) (11.7.3
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in /usr/local/lib/python3.12/dist-packages (from torch) (12.5.8
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2.27.5)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch) (3.4.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch) (12.8.90)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch) (12.8.93
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.13.1.3)
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings==12.9.4->to
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformer
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->data
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.11
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.(
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.(
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspe
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hul
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->hug
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->datase
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.(
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.(
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.(
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->hug
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->hu
```

```
import torch
from datasets import load_dataset
from transformers import AutoTokenizer, AutoModelForSequenceClassification
from transformers import TrainingArguments, Trainer
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
```

```
dataset = load_dataset("glue", "sst2")
print(dataset)
```

```
DatasetDict({
    train: Dataset({
        features: ['sentence', 'label', 'idx'],
        num_rows: 67349
    })
    validation: Dataset({
        features: ['sentence', 'label', 'idx'],
        num_rows: 872
    })
    test: Dataset({
        features: ['sentence', 'label', 'idx'],
        num_rows: 1821
    })
})
```

```
tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")
```

```
def tokenize_function(example):
    return tokenizer(example["sentence"], padding="max_length", truncation=True)

tokenized_datasets = dataset.map(tokenize_function, batched=True)
```

Map: 100% 872/872 [00:00<00:00, 1701.40 examples/s]

```
train_dataset = tokenized_datasets["train"].select(range(5000))
test_dataset = tokenized_datasets["validation"].select(range(tokenized_datasets["validation"].num_rows))
```

```
train_dataset = train_dataset.remove_columns(["sentence","idx"])
test_dataset = test_dataset.remove_columns(["sentence","idx"])
```

```
train_dataset.set_format("torch")
test_dataset.set_format("torch")
```

```
model = AutoModelForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=2
)
```

model.safetensors: 100% 268M/268M [00:02<00:00, 250MB/s]

Loading weights: 100% 100/100 [00:00<00:00, 171.09it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]

```
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                     | Status     |
------------------------+------------+-
vocab_transform.weight  | UNEXPECTED |
vocab_layer_norm.bias   | UNEXPECTED |
vocab_layer_norm.weight | UNEXPECTED |
vocab_projector.bias    | UNEXPECTED |
vocab_transform.bias    | UNEXPECTED |
pre_classifier.bias     | MISSING    |
classifier.bias         | MISSING    |
classifier.weight       | MISSING    |
pre_classifier.weight   | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
```

```
def compute_metrics(pred):

    labels = pred.label_ids
    preds = pred.predictions.argmax(-1)

    precision, recall, f1, _ = precision_recall_fscore_support(
        labels, preds, average='binary'
    )

    acc = accuracy_score(labels, preds)

    return {
        "accuracy": acc,
        "f1": f1,
        "precision": precision,
        "recall": recall
    }
```

```
training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=1
)
```

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset,
    compute_metrics=compute_metrics
)
```

```
trainer.train()
```

[313/313 03:49, Epoch 1/1]

Step	Training Loss

Writing model shards: 100% 1/1 [00:06<00:00, 6.09s/it]

```
TrainOutput(global_step=313, training_loss=0.3827663580068765, metrics={'train_runtime': 232.1859, 'train_samples_per_second': 21.534, 'train_steps_per_second': 1.348, 'total_flos': 662336993280000.0, 'train_loss': 0.3827663580068765, 'epoch': 1.0})
```

```
trainer.evaluate()
```

[55/55 00:13]

```
{'eval_loss': 0.31290459632873535,
 'eval_accuracy': 0.8715596330275229,
 'eval_f1': 0.8766519823788547,
 'eval_precision': 0.8577586206896551,
 'eval_recall': 0.8963963963963963,
 'eval_runtime': 13.844,
 'eval_samples_per_second': 62.988,
 'eval_steps_per_second': 3.973,
 'epoch': 1.0}
```

```
text = "This movie was amazing and very interesting!"

inputs = tokenizer(text, return_tensors="pt")

# Move input tensors to the same device as the model
inputs = {k: v.to(model.device) for k, v in inputs.items()}

outputs = model(**inputs)

prediction = torch.argmax(outputs.logits)

if prediction == 1:
    print("Positive Sentiment")
else:
    print("Negative Sentiment")
```

```
Positive Sentiment
```

```
import matplotlib.pyplot as plt

logs = trainer.state.log_history
loss = [x["loss"] for x in logs if "loss" in x]

plt.plot(loss)
plt.xlabel("Steps")
plt.ylabel("Loss")
plt.title("Training Loss Curve")
plt.show()
```